

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Mid-Term Examination, Spring 2025
Course Code: CSE 3521 Course Title: Computer Architecture
Total marks: 30 Time: 1.5 hours

- | | | | |
|---|-----------|----------|---------|
| 1. a) Define <i>CPI</i> and explain how it affects CPU performance. Explain with example | Mark
3 | CLO
2 | DL
2 |
| ↓ b) Why is MIPS not always a reliable performance measure? Explain with example | 3 | 2 | 2 |
| 1. c) Different instruction sequences can yield different execution times for the same task. Given two compilers A and B, both targeting the same hardware: | 4 | 2 | 3 |

Class	CPI	A Count	B Count
A	1	5	4
B	2	2	3
C	3	1	2

Assuming a 2.5 GHz clock, compute which compiler produces a faster code and justify your answer with proper reasoning.

- OR
- Consider the following two processor designs for the same ISA:
Processor A: Clock Rate = 2.5 GHz, Average CPI = 1.6
Processor B: Clock Rate = 2.0 GHz, Average CPI = 1.2
 A program takes 1.5 billion instructions to execute.
 a. Compute the execution time on each processor.
 b. Which processor is faster and by what percentage?
 c. If performance is critical, which design would you recommend?

Handwritten calculations for Processor A and B:

Processor A: $\text{Execution Time} = \frac{\text{Instructions}}{\text{Clock Rate} \times \text{CPI}} = \frac{1.5 \times 10^9}{2.5 \times 10^9 \times 1.6} = 0.375 \text{ s}$

Processor B: $\text{Execution Time} = \frac{\text{Instructions}}{\text{Clock Rate} \times \text{CPI}} = \frac{1.5 \times 10^9}{2.0 \times 10^9 \times 1.2} = 0.625 \text{ s}$

Processor A is faster.

2. a) Translate the following MIPS instructions into machine code:
 (i) `addi $t1, $t2, 15`
 (ii) `beq $s1, $s2, -8`
 (iii) `lw $t0, 32($sp)`

- OR
- Translate the following machine code into MIPS instructions:
 001100 01011 01010 0000000011110000
 101011 10001 10000 111111111101100
 000101 00101 00110 0000000000000100

- | | | | |
|--|---|---|---|
| 2. b) Describe the purpose and structure of an instruction format. How it aligns with design principles. | 4 | 1 | 2 |
| 3. a) A computer system is using the IEEE 754 format to store floating-point numbers. Why is normalization important in this format? Show how the number -5.75 is normalized and represented in single precision. | 6 | 1 | 3 |
| 3. b) Why refined versions of multiplication hardware reduce its size and components? Explain with examples. | 4 | 1 | 2 |

International Islamic University Chittagong
Department of Computer Science & Engineering
Program: B. Sc (Engg.) in CSE Semester: 5th
Mid-Term Examination, Spring 2024

Course Code: CSE-3521

Course Title: Computer Architecture

Time: 1 hour and 30 minutes

Total Marks: 30

[Parts of the same question must be answered following the serial]

- Q.1.** a. Define the terms CPI and CPU clock cycle. "MIPS can vary inversely with performance"- verify this statement with appropriate example.
 b. Suppose we have two implementations of the same instruction set architecture. Machine A has a clock cycle time of 1 ns and a CPI of 3.0 for some program, and machine B has a clock cycle time of 2 ns and a CPI of 2.2 for the same program. Which machine is faster for this program, and by how much?

Marks CLO DL

4 1 1

4 2 3

1 x clock cycle time

Or

Codes	Instruction Counts for instruction class(in billions)		
	A	B	C
Compiler 1	4	2	3
Compiler 2	7	6	5

Instruction Class	CPI for this Class
A	1
B	2
C	3

Assume that the computer's clock rate is 5 GHz. Which code sequence will execute faster according to MIPS? According to execution time?

- c. Subtract (35-7) using 2's complement.

- Q.2.** a. Define instructions. What is the major difference between the variables of a programming language and registers?
 b. "Good Design Demands Good Compromises"-is one of the design principles of MIPS. Give proper explanations with example.
 c. Translate the following MIPS Assembly Language instructions into Machine Language. You must specify the MIPS fields and Instruction Format used.
 (i) add \$s1, \$t5, \$t6.
 (ii) bne \$s2, \$s4, 100
 (iii) sw \$s1, 100(\$s2)
 d. What are the reasons for considering 32 bit registers each of size 32? Explain

1+1=2 1 1

3 1 2

3 2 3

*So - to 6-15
 16-23 to 8
 24-31 to 0
 17 12 11 10 9 8 7 6 5 4 3 2 1 0*

Or

What is spilling register? When it is necessary to spill register?

- Q.3** a. Briefly describe different addressing modes in MIPS.
 b. Show the IEEE 754 binary representation of the number -0.75_{10} in single and double precision.

2 1 1

4 2 3

Or

Describe the first version of multiplication hardware using flowchart.

- c. Multiply $4_{10} \times 5_{10}$ using final version of multiplication algorithm.

4 2 3

11011

*11011 1010100
 01011 101011
 1100 11*

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science and Engineering

Mid-Term Examination, Autumn-2023

Course: CSE-3521 (Computer Architecture)

Time: 1 hour and 30 Minutes

Marks: 30

[Answer all *three* questions; Figures in the right hand margin indicate Marks]

- Q1. a) Define Computer Architecture. Explain the subcategories of computer architecture 1+2 CO1
- b) A program runs in 10 seconds on computer A, which has a 300 MHz clock. We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program. What clock rate should we tell the designer to target? 4 CO2
- c) Suppose we have two implementations of the same instruction set architecture. Computer1 has a CPI of 2.0 and a clock cycle time of 250 ps for some programs, and computer 2 has a CPI of 1.2 and a clock cycle time of 500 ps for same programs. Which computer is faster for this program and by how much? 3 CO2
- Q2. a) State four design principles of MIPS 2 CO1
- b) Translate the following MIPS Assembly Language instructions into Machine L 4 CO2
- (i) sub \$s2, \$t5, 20
 - (ii) bne \$s3, \$s1, 100
 - (iii) sw \$s7, 80(\$s2)
 - (iv) slt \$s4, \$s6, \$s7

c) What is the difference between CISC and RISC computers? 2 CO1
Or
Which registers are reserved for assembler and operating system?

d) Describe the MIPS addressing mode with figure. 2 CO1

Q3. a) Describe the first version of division Algorithm using Flow Chart. 3 CO1

Or

Describe the Final version of Multiplication Algorithm using Flow Chart.

b) Show the IEEE 754 binary representation of the number -0.25_{10} in single and double precision 3 CO2

c) Divide $5_{10} \div 3_{10}$ using final version of ^{division} ~~multiplication~~ algorithm 4 CO2

$$\begin{array}{r} 1 \\ 3 \overline{) 5} \\ \underline{3} \\ 2 \end{array}$$

Bismillahir Rahmannir Rahim
International Islamic University Chittagong
Department of Computer Science and Engineering
Mid-Term Examination, Spring-2023
Course: **CSE-3521 (Computer Architecture)**
Time: **1 hour and 30 Minutes** Marks: **30**

[Answer all **three** questions; Figures in the right hand margin indicate Marks]

Q1. a) Define the following:

2 **CO1**

(i) Computer Architecture (ii) CPU time

b)

Code Form	Instruction Counts for instruction class (in billions)		
	A	B	C
Compiler 1	4	2	3
Compiler 2	10	5	Last digit of your student ID

3 **CO2**

Instruction Class	CPI for this class
A	1
B	2
C	3

Assume that the computer's clock rate is 5 GHz. Which code sequence Will execute faster according to MIPS according to execution time?

c) Suppose we have two implementations of the same instruction set architecture. Computer1 has a CPI of 3.0 and a clock cycle time of 300 ps for some programs, and computer2 has a CPI of 4.0 and a clock cycle time of 150 ps for same programs. Which computer is faster for this program and by how much?

3 **CO2**

d) What is CPU clock cycle and clock rate? UNIX time command is given 40.Xu 10.5s 2:50 calculate the percentage according to total elapsed time. [N.B: X=last digit of your student ID]

2 **CO2**

Q2. a) Explain different types of instruction format.

2 **CO1**

- b) Translate the following MIPS Assembly Language instructions into Machine Language. You must specify the MIPS fields and Instruction Format used. 4 CC
- (i) add \$s5, \$t5, 20
 - (ii) bne \$s3, \$s1, 100
 - (iii) sw \$s7, 80(\$s2)
 - (iv) beq \$s1, \$s6, 24
- c) What is the difference between CISC and RISC computers? 2 CO1
- (d) Describe the MIPS addressing mode with figure. 2 CO1
- Q3. a) Describe the first version of division Algorithm using Flow Chart. 3 CO1
- Or
- Describe the Final version of Multiplication Algorithm using Flow Chart.
- b) Show the IEEE 754 binary representation of the number -0.375_{10} in single and double precision 3 CO2
- c) Multiply $2_{10} \times 5_{10}$ using final version of multiplication algorithm 4 CO2

[Answer all the questions. Figures in the right margin indicate full marks.]

- 3.
- | | | | | |
|----|--|---|-----|----|
| a) | Mention the <i>registers</i> used in <i>read</i> operation from memory (RAM) and how? | 3 | CO2 | C1 |
| b) | Write the differences between Indirect, Direct addressing mode with example. | 2 | CO2 | C2 |
| c) | How <i>relative</i> addressing mode works? Show example. | 1 | CO2 | C1 |
| d) | Write a subroutine, SEARCH, to search for a value VAL in a list of N values and write the position as an output. | 4 | CO2 | C2 |
- OR
- | | | | | |
|----|---|---|-----|----|
| d) | Draw the hardware design of second version Multiplication Algorithm and describe using flowchart. | 4 | CO2 | C2 |
|----|---|---|-----|----|

END